Roberto Gallego Barbarán

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EDUCATION AND SPECIALIZED TRAINING

University of Málaga 2022 – 2026
BSc in Software Engineering Málaga, Spain

University Extension in Blockchain Technologies (490h) Feb. 2026 – Jun. 2026
University of Málaga (UMA) Málaga, Spain

- Distributed systems, smart contracts (Solidity, Daml), decentralized infrastructure (Canton, gRPC), and blockchain audit fundamentals.

EXPERIENCE

DEKRA Sept. 2026 – Mar. 2027
AI & Automation Engineer Intern Málaga, Spain

- Automated laboratory-results analysis by building AI agents with Python, LangChain and LangGraph, implementing RAG architectures with vector databases that reduced manual processing time and improved technical-data lookup accuracy.
- Designed intelligent automation pipelines integrating LLMs with external tools via MCP (Model Context Protocol), orchestrating state graphs with LangGraph for complex decision flows and removing repetitive internal tasks.

IOBuilders – University-Industry Collaboration 2026
Backend & Blockchain Developer Málaga, Spain

- Delivered a tokenized bond platform by designing the Go backend and Daml contracts, orchestrating multi-service infrastructure with Docker Compose and Nginx, and validating issuance, transfer, and burn flows with Playwright end-to-end tests under IOBuilders coordination.
- Ensured transactional consistency between Go business logic and the Canton Ledger via gRPC, validating every distributed flow with automated tests covering the full token lifecycle.

PROJECTS

Intelligent Game Validation | *TypeScript, Node.js, Playwright, OCR, GitHub Actions* | GitHub 2026

- Placed 4th at Malackathon 2026 (University of Málaga hackathon, Betsson challenge) in a team of four with a TypeScript/Playwright tool that validates third-party casino games inside cross-origin canvas iframes as the player sees them, with no DOM access.
- Engineered a hybrid computer-vision pipeline (pixelmatch frame diffing, masked template matching, tesseract.js OCR), cross-checked with intercepted provider traffic, to verify spin completion, wager, currency and balance reconciliation.
- Automated CI in GitHub Actions with type checking, 42 unit tests, a live run on the public demo game and a fault-injection negative control that must fail, publishing screenshots, video and HTML reports.

Bancosol Campaigns | *Java 17, Spring Boot, Spring MVC, JPA, Spring Security, JWT, SQL* | GitHub 2026

- Architected the campaign management module with a layered architecture (Spring MVC, services, JPA repositories, DTOs/mappers) and RBAC via Spring Security and JWT, enforcing multi-tenant data isolation.
- Ensured data integrity and session-based filtering by implementing the persistence layer on MariaDB/MySQL for the full domain model of stores, shifts, volunteers, assignments, and incidents, validated with JUnit and Mockito.

Cava Bot | *Python, GitHub Actions, Gemini API, yfinance, NewsAPI, Telegram API* | GitHub 2026

- Eliminated dedicated server dependency for a daily financial notification bot by designing a serverless pipeline with GitHub Actions and cron, orchestrating external APIs (Gemini, yfinance, NewsAPI) through secrets and environment variables.
- Automated full deployment and daily execution via CI/CD with GitHub Actions, reducing operational cost to on-demand external API consumption with no persistent infrastructure.

TECHNICAL SKILLS

AI & Machine Learning: LangChain, LangGraph, MCP, RAG, vector databases, LLMs, AI agents, Gemini API, prompt engineering, computer vision/OCR.
Backend & APIs: Java 17, Spring Boot, Spring MVC, JPA, Spring Security/JWT, Maven, SQL, MariaDB/MySQL, REST APIs.
DevOps & Infrastructure: Docker, Docker Compose, Kubernetes, GitHub Actions, Linux, Nginx, CI/CD, Terraform/IaC fundamentals.
Languages & tools: Python, Go, TypeScript, JavaScript, C/C++, Git, JUnit/Mockito, Playwright.
Blockchain & distributed systems: Solidity, Daml, Canton, Smart Contracts, gRPC.
Spoken languages: Spanish native, English C1.